LINH (LILIAN) TRAN (SHE/HER)

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EDUCATION

San Jose State University *Graduating May 2026*
Master of Engineering, Software Engineering
Cohort: Cloud Engineering

University of California, Berkeley *Graduated May 2023*
Bachelor of Arts, Computer Science and Data Science

Awards and Honors: Member of Upsilon Pi Epsilon, the Computer Science international honor society.

SKILLS

- Python, C, C++, C#, Golang, R, Java, JavaScript, TypeScript, SQL, NoSQL
- Linux, Windows, iOS, Git, Docker & Kubernetes, AWS, GCP, REST/RESTful API, Fast API, RDBMS, Jira, Jenkin, NLP.
- TensorFlow, PyTorch, scikit-learn, OOP, OOD, Big Data, Machine Learning.

Having solved 600+ Leetcode problems, with over 55% of them Medium/Hard level.

RELEVANT EXPERIENCE

Saigon Fabrication *San Jose, CA*
Software Engineer Nov 2023 – Present

- Designed and developed application for extracting, processing, analyzing and storing hardware test results for future reference.
- Trained and applied AI models to increase the effectiveness of hardware components identification and classification.
- Analyzed hardware schematics and designed hands-on tests and measurement standards to ensure the quality of the final hardware products.

The University of Texas Health Houston *Houston, TX*
AI Research Assistant Jan 2023 – Oct 2023

- Leveraged the power of GPU in Jupyter Notebook to accelerate the ETL process and Machine Learning tasks in the local host.
- Built an NLP ML model implementing text vectorization and the LSTM, Transformation Model to process 69,832 ICD-10 codes and group them into categories based on their descriptions for healthcare research purposes.
- Implemented PCA, LSA, PLSA and LDA dimensionality reduction techniques on data points with over 1000 dimensions and used their output as training/testing input for convolutional neural network models, analyzed and visualized the outcomes using t-SNE.

UC Berkeley Electrical Engineering & Computer Science Department *Berkeley, CA*
Software Engineer Intern Aug 2022 – Dec 2022

- Developed a simplified version of Google Drive featuring password-protected accounts utilizing the HMAC technique for password validation and storing specific secret keys which authorized access and modification to AES-CBC encrypted files.
- Created a secured key-sharing channel using RSA key exchange methods and built one tree for each encrypted file to keep track of the file's owner, users received permission to that file and their secondary recipients, which allowed instant permission revocation.
- Utilized VM to operate the whole system; defined a document to contain information from files and stored them in MongoDB.

TMA Solutions *Ho Chi Minh, Vietnam*
AI Engineer Intern May 2022 – Aug 2022

- Configured the GPU setting and CI/CD pipeline to create a suitable environment for training and testing large-scale data.
- Analyzed the classification system thoroughly and designed a compatible data preprocessing program to sharpen and augment the low-quality inputs of approximately 2 million data points prior to feeding those inputs to the classification system.
- Collaborated with team members to nominate the best classification model and enhanced it to achieve a 5% precision improvement compared to the previous model.

UC Berkeley Electrical Engineering & Computer Science Department *Berkeley, CA*
Course Staff Jul 2020 – Dec 2020

- Reinvented course delivery, focusing on problem-solving strategies for a class of 1000+ students, increased total exam score by 5%.
- Conducted over 40 informative office hours to address students' inquiries on various topics, including object-oriented programming, algorithms, and data structures through several communication platforms such as Zoom, Google Meet, Slack, Piazza, etc.
- Collaborated with professors and 130 other course staff to enhance course materials for improved student comprehension.

De Anza College *Cupertino, CA*
Teacher Assistant Jul 2018 – Jun 2019

- Walked students through steps to debug and troubleshoot their code, empowering them to identify and resolve issues independently.
- Delivered comprehensive explanations of programming concepts to strengthen students' foundational knowledge.
- Helped them understand the core concept of divide and conquer in computer science.

Math Tutor Sep 2018 – Jun 2019

- Held office hours to resolve students' concerns and answer questions related to course materials such as calculus, differential equations, linear algebra, etc.
- Taught one-on-one tutorial sessions to help students catch up with their coursework by using methods that were tailored individually.

LEADERSHIP & ACTIVITIES

De Anza Associated Student Government, *Senator* *Nov 2018 – Jun 2019*

- Collaborated with fellow senators to allocate a budget exceeding $1 million to fund various campus programs and organizations like EOPS, Student Success Center, La Voz News, campus clubs, and student athletes.

Competitive Programming Club, *Internal Vice President* *Sep 2018 – Jun 2019*

- Cultivated a close-knit and competitive community for students interested in coding challenges and competitions.
- Established a tightly knit community for students passionate about tackling complex coding challenges.

Rank 8th in the North California Site/Top 30 in the Pacific Northwest Region in the ACM International Collegiate Programming Contest, Division II.